

ASMRFeel: A Multi-Modal Thermal-Haptic Feedback System for Live ASMR Interaction

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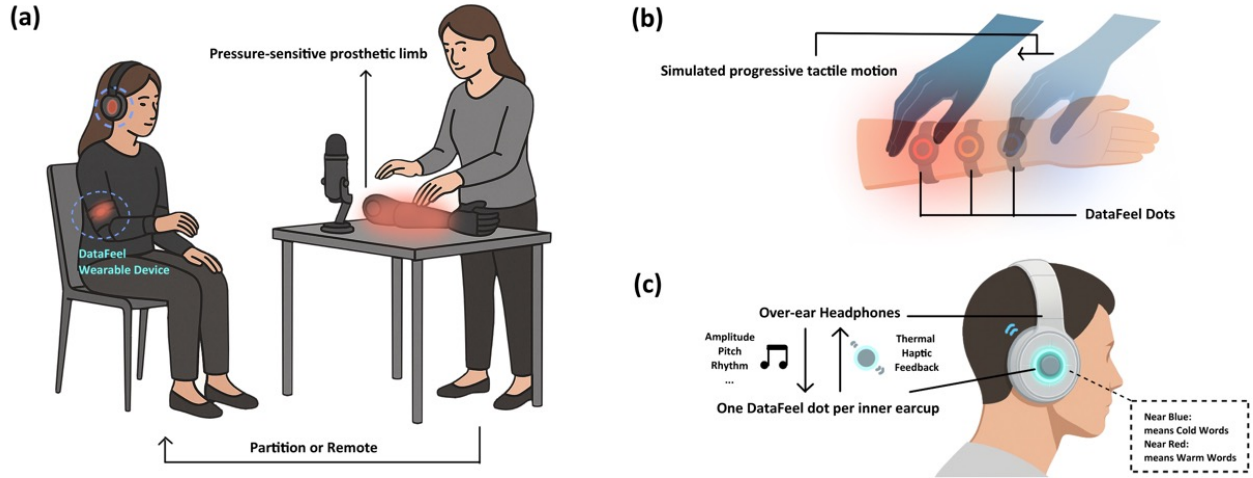


Figure 1: Illustration of ASMRFeel system components and interaction flow. (a) The viewer wears thermal-haptic-enabled DataFeel devices and over-ear headphones, while the ASMRtist interacts with a pressure-sensitive prosthetic limb. A microphone captures the ASMRtist's speech and ambient audio. (b) Linear thermal and vibrotactile sensations are synthesized by sequentially activating DataFeel Dots, simulating progressive stroking along the user's forearm. (c) Auditory features such as amplitude, pitch, and rhythm are translated into synchronized thermal-haptic feedback at the earcups. Color and temperature cues (e.g., blue for cold-toned words, red for warm-toned words) enhance emotional immersion.

Abstract

ASMRFeel is a multi-modal feedback system that enhances ASMR live streaming by integrating spatially distributed thermal and vibrotactile feedback. Utilizing multiple DataFeel Dot actuators placed on the viewer's body, ASMRFeel enables the perception such as linear stroking-like sensations by modulating temperature and haptic feedback over time and space. The system captures touch information such as force, along with thermal data within the context in the narrative performed by ASMRtist via a pressure-sensitive surface, transmits corresponding feedback to remote users, thereby extending traditional ASMR's auditory focus into tactile domain. This work demonstrates stimulation in real time, enabling immersive, pseudo-physical interaction between ASMRtist and viewers. We

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discuss the system architecture, thermal interpolation model and perceptual considerations for rendering continuous, human-like tactile experiences remotely.

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1 Introduction

ASMR, or Autonomous Sensory Meridian Response, has gain increasing attention in recent years. ASMRtist perform ASMR through video or audio recording, allowing viewers to receive tingling sensations or stimuli, which help relieve depression, stress and chronic pain[2, 3]. Prior research from Peng et al.[5] introduce vibrotactile feedback sensation when experiencing ASMR in Virtual Reality environment, the result shows that vibrotactile feedback enhanced user perceptions of enjoyment. Apart from tactile haptic such as

vibrotactile feedback, thermal stimulation and visual cues such as LED are other commonly employed modality of haptic feedback. Studies have shown that using thermal stimulation[1, 4] or visual cues[6] can influence emotional states.

In this work, we present ASMRFeel, a multimodal system that integrates vibrotactile feedback, thermal stimulation, and LED visual cues to enhance the ASMR experience for viewers. ASMRFeel also incorporates input modalities for ASMRtist enabling responsive haptic content generation.

With the integrated haptic feedback, viewers can experience ASMR in a more immersive manner. For example, they can feel the virtual touch performed by the ASMRtist, perceive the real-time body temperature of a character within the narrative, the LED can further act as a visual cue guiding the viewer's attention or to create a ambient aura that complements the narrative atmosphere, enriching the sensory realism of the experience and deeper engage with the narrative.

2 Methodology

2.1 ASMRtist : Gesture Acquisition and Encoding

During a live session, the ASMRtist simultaneously generates soft auditory triggers (whispers, brushing sounds) and tactile gestures on two sensing zones: (i) a pressure-sensitive prosthetic forearm equipped with two capacitive sensors, and (ii) a pair of miniature capacitive sensors mounted near each ear to capture interactions such as cupping or tapping around the pinna. While the audio stream is captured and transmitted conventionally, all sensors sample forces in real time, flag meaningful contact, determine which sensor is dominant to pinpoint the touch location, and classify events as taps, presses, or strokes via temporal analysis. Linear interpolation between the forearm and ear sensors produces a unified one-dimensional path, and every frame is time-stamped so the resulting stream of position, force, and gesture type aligns precisely with the audio feed for synchronized playback.

2.2 Viewer : Haptic and Thermal Rendering

Viewers use a hybrid haptic setup: two DataFeel Dot actuators are embedded in the ear-cups of their headphones, while the remaining two are worn on the forearm. For each gesture packet, the rendering engine smoothly modulates vibration and temperature across this combined array, producing a single coherent sensation that can glide from the arm to the region around each ear. Tap labels trigger brief vibration bursts and quick temperature pulses; presses maintain steady output; strokes generate sliding envelopes—all timed to the accompanying audio. Thermal set-points stay within comfort limits, delivering synchronized vibrotactile and thermal cues that envelop both the forearm and the ears, faithfully reproducing the ASMRtist's sound-and-touch performance as an immersive, full-body experience.

3 Implementation

3.1 Signal Pipeline

To ensure low latency in ASMRFeel, we design a pipeline that processes both tactile gestures and voice signals in parallel. For the

tactile path, the 2D pressure map is locally processed into a discrete format comprising segmented regions. In parallel, the *Voice-to-Feel* (V2F) module extracts key audio features from the ASMRtist's voice stream, especially amplitude and pitch. Finally, a shared control buffer ensures synchronization across auditory, thermal, and haptic streams.

3.2 Mapping Pressure Array to Single-Point Actuators

The key challenge arises from the transition between the pressure-sensitive surface and the sequential single-point DataFeel Dots. To resolve this, we design the mapping process involves the following steps:

- **Gaussian Smoothing and Region Clustering.** At each time step, the 2D pressure map is smoothed using a Gaussian kernel to reduce high-frequency noise and emphasize coherent contact regions. Clustering (e.g., k-means or DBSCAN) is then applied to segment the pressure map into discrete regions corresponding to the locations where DataFeel Dots are worn.
- **Computing the Weighted Centroids.** For each clustered pressure region, a weighted centroid is computed based on pressure intensity values.
- **Signal Aggregation and Actuator Encoding.** The weighted centroid within each mapped region is then mapped to both vibration amplitude and thermal intensity using different transformation functions. These values are encoded and transmitted to the respective DataFeel Dots later.

3.3 Speech-to-Tactile and Thermal Conversion

To enrich the ASMR experience beyond traditional voice-based interactions, we developed a *Voice-to-Feel* (V2F) module that converts audio features into tactile and thermal stimuli:

- **Amplitude (volume)** maps to vibration intensity.
- **Pitch (frequency)** maps to thermal transition.
- **Temporal texture** such as rhythm and density modulate the vibration pattern.

In addition, a lightweight audio classification model detects common ASMR triggers (e.g., tapping, whispering, licking, or sigh sounds), which can activate predefined haptic or thermal feedback sequences.

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